



Our All-Purpose Solution

AR 403

- Ideal for R&D and small scale production in the pharmaceutical, chemical, cosmetic and food industry
- For pharmacy, university and laboratory applications
- Modular design – one drive unit and various attachments for different uses and needs

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AR 403



The ERWEKA All-Purpose System is based on a powerful drive unit available in two versions. The various attachments can be easily and quickly connected by means of a coupling.

The drive unit AR 403 is equipped with a three-phase AC motor.

The sturdy ball bearing worm gear drive is completely maintenance-free and provides an output torque of max 30 Nm. The unit features electronic speed control variable between 0 and 400 rpm. A digital dial regulates the speed.

Through permanent nominal/actual adjustment the preselected speed is kept, even under varying load.

The AR 403-S is similar to the AR 403 but has a different drive train. This connection is designed for heavy use.

Available options:

- + Foot switch
- + USB interface for printout
- + IQ / OQ / PV documents

Ideal for R&D and small scale production

The All-Purpose equipment is ideal for small scale production in the pharmaceutical, chemical, cosmetic and food industries. It fits perfectly into Research & Development of tablets, ointments, creams and more.

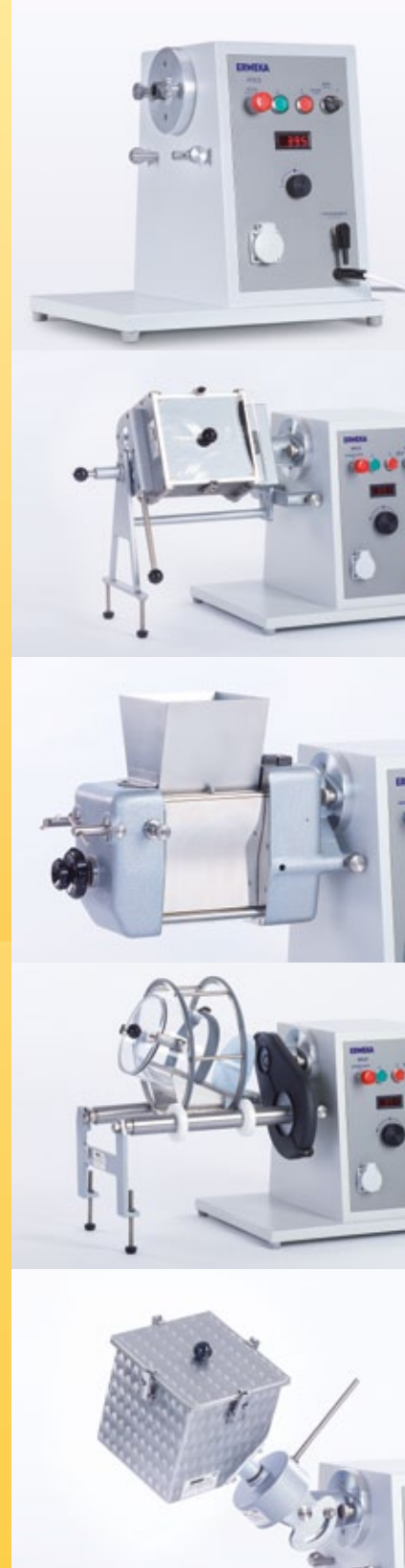
The modular design is compact, economical and offers a powerful motor drive with a wide range of easily interchangeable attachments for various purposes.

The different possibilities are as follows:

- Stirring
- Kneading
- Mixing
- Granulating
- Coating
- Polishing
- Grinding
- Sieving
- Homogenizing
- Filling
- Dosing
- Tablet production

Technical Specifications

	AR 403 / AR 403-S
Torque	Robust / -S for heavy use
Voltage/frequency	230V 50/60 Hz 115 V/60 Hz 127V/60Hz
Width	450 mm
Length	400 mm
Height	440 mm
Weight	30 kg
Motor type	Three-phase AC motor
Power	750 W
Tropics insulation	Class F
RPM	20-400 rpm
Tolerance rpm (coasting)	+/- 1 rpm
Gear type	Worm gear
Output	20-400 rpm
Gear max. torque	23 Nm
Efficiency	89%
Timer	
Possible time setting	Hours/minutes
Set time (max)	9h 59min
Tolerance timer	+/- 1 sec



Main benefits of the modular ERWEKA method:

- Little lab space needed
- Constant product quality - improved production
- Economical capital cost - one drive, many attachments
- High quality - made in Germany

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AR 403 Attachments

ERWEKA UG (Universal Gear)



The universal gear is required for adjusting the angle of operation (0-80°) and for speed reduction (1:6).

The UG is required for the operation of

- Ball mills (KM 0)
- Cube mixer (KB 15/20/S)
- Double Cone Mixer (DKM)
- Pelletizer (GTE)
- Coating pan (DKE/S)
- Polishing drum (PT)

ERWEKA KM (Ball Mill)



Ball mill for grinding crystalline materials and for mixing dry, or under certain circumstances, moist materials.

The balls are made of porcelain having a diameter from 30 to 50 mm. The drum is sealed airtight with a lid and is delivered including frame for drive unit. Drum and balls can be sterilized in water. For using the ball mill, a drive support is needed, which is optionally

available lengthened for the simultaneous use of two mixers of the same size (KM 5 l or KM 10 l volume, working capacity max. 40%).

The ball mill KM 0 (with 0,5 l volume, working capacity of max. 40% and made of stainless steel) is attached to the drive unit via the ERWEKA Universal Gear. The drum is available with 0,5, 5 or 10 l volume.

ERWEKA CM (Conical Mill)



The Conical Mill is a product infeed system for application in developments in the pharmaceutical and food industries, which pulverizes powdery and free flowing products by a sieve with fixed maximum sizes of impurity-free granule (adjustable 0.3 – 3.0 mm; perforated round holes, grater holes or square holes).

Rotation speed (continuous 60 – 1200 rpm) and sieve crack (standard 1.5 mm, adjustable 0.5 – 3.0 mm in 0.25 mm steps) can be varied for detailed adjustment of process.

ERWEKA SM/SMS (Three Roller Mill)



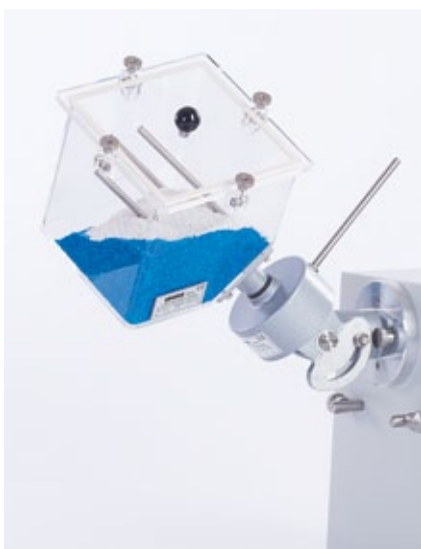
Three roller mill for the production of pastes, ointments, creams and similar products.

The rollers are manufactured in porcelain (SM) or stainless steel (SM S). Each roller has a diameter of 50 mm.

The rollers are spring loaded to prevent damage and the space between the rollers is adjustable by one screw.

The mill has an output between 2 kg/h and 15 kg/h depending on the consistency of the product to be produced.

ERWEKA KB (Cube Mixer)



The cube mixer uses a tumbling motion to produce a homogenous blend within a short time. The mixing action is assisted by three stainless steel rods strategically positioned within the cube.

The cube mixer is attached to the drive unit via the ERWEKA Universal Gear UG, which permits operation at various angles for optimum efficiency.

Our cube mixers are available in the following versions:

- **(KB 15)** Cube mixer made of acrylic glass with a volume of 3.5 l and 40% working capacity

- **(KB 15 S)** Cube mixer made of stainless steel, 3.5 l volume, 40% working capacity
- **(KB 20)** Cube mixer made of acrylic glass, 8 l volume, 40% working capacity
- **(KB 20 S)** Cube mixer made of stainless steel, 8 l volume, 40% working capacity
- **(KB 30)** Cube mixer made of acrylic glass, 11 l volume, 40 % working capacity
- **(KB 30 S)** Cube mixer made of stainless steel, 11 l volume, 40 % working capacity

ERWEKA DKM (Double Cone Mixer)



Double cone mixer for the mixing of free flowing powders and granules.

The arrangement of the mixer in combination with the angle adjustment of the universal gear allows a multi-dimensional mixing process.

The mixing vessel (11.9 liter volume, working capacity 50%) is made of stainless steel.

The ERWEKA Universal Gear UG is needed for connection and allows operation at various angles for optimum efficiency.

AR 403 Attachments

ERWEKA RM (Drum Hoop Mixer)



Drum hoop mixer with diagonal placement of the drum within the hoop. The rotation of the hoop itself produces a uniform, three-dimensional displacement of the material to be processed and hence a tumbling mixing action.

The mixer consists of two parts: the drive support and drum hoop with acrylic mixing drum. For using the drum hoop mixer a drive

support is needed, which is optionally available lengthened for the simultaneous use of two mixers of the same size (RM 5 or RM 10 liter volume, working capacity max. 40%).

Mixing drums are available with 5 liter and 10 liter volumes. Ball mill drums are available as optional accessories.

ERWEKA SW 1/S (Laboratory High Speed Mixer)



Laboratory high speed mixer operating on the plough-share principle for the mixing of powders, granules and other products of fibrous nature.

Quantity ratios of 1:100,000 can be mixed homogeneously in only a few minutes. The mixing drum (5 liter volume, working capacity max. 70%) can be rotated through 180 degrees and offers a removable cover for charging

and emptying. An electro-mechanical cover switch ensures that the drive unit AR 403 is switched off when the cover is removed.

All parts coming into contact with the product are made of stainless steel.

ERWEKA Y5 (Mixer Y5)



Mixer Y5 with Y-shaped mixing vessel for the gentle mixture of solid, well flowing materials (bulk materials).

The rotary motion of the mixing vessel makes a particularly homogenous mixture possible and suitable for sensitive materials.

For optimal filling, emptying and cleaning the mixing vessel is equipped with three

capacious openings. Volume of the mixing container: 5 liter, recommended working conditions: max. 1.5 liter (30%) and 0-40 RPM.

All parts coming into contact with the material are made of stainless steel.

ERWEKA PRS (Planetary Stirrer)



Planetary stirring unit for mixing of creams, ointments, pastes, moist powders and liquids.

This powerful mixing attachment comes with various mixing/stirring accessories e.g. kneading hook, mixing paddle, whisk and a fluoropolymer scraper for removing any product adhering to the side of the bowl thus ensuring a homogenous mixing in the shortest possible time.

The stirring/mixing accessories and vessel (5 liter volume, working capacity 60%) are made of stainless steel.

Optionally an electric heated bowl or a double jacket bowl for heating and cooling by water is available.

ERWEKA LK 5 (Laboratory Kneader)



Laboratory kneader with stainless steel kneading vessel (5 liter volume, working capacity 65%), housing, two z-shaped kneading blades running at different speeds in opposing directions.

The asynchronous mixing action intensifies and considerably accelerates the kneading process. The kneading vessel is fitted with a plexiglas cover to monitor the kneading

process. An electro-magnetic cover lock ensures that the power supply to the drive is disconnected immediately when the vessel cover is removed.

A thermostatically controlled heating jacket for the kneading vessel is available as an optional extra.

ERWEKA FGS (Wet Granulator)



Wet granulator with oscillating rotor for the production of wet granules.

The granules can be produced in varying sizes depending on the sieve mesh employed.

All parts coming into contact with the product are made of stainless steel. Working capacity is max. 30 kg/h depending on the material

and mesh width of the sieve used. The unit is supplied as standard with two sieves 1.0 mm and 1.6 mm MW respectively.

The following mesh sizes are also available: 0.315; 0.63; 0.8; 1.0; 1.25; 1.6; 2.0; 2.5 and 3.15 mm. Other special sieves are available on request.

ERWEKA TG 2000 (Dry Granulator)



Dry granulator with two driven rollers running in opposing directions from the inside to the outside.

The rollers are 70 mm long and made of stainless steel. They are equipped with teeth to produce a powerful grinding action by forcing the material against a series of braking combs. The ERWEKA TG 2000 fully conforms to the latest safety requirements.

All parts coming into contact with the product are made of stainless steel. Fines produced during granulation are very low (max. 25%) and can be reduced to 15% by careful adjustment of the process.

The maximum capacity of the dry granulator is 20 kg/hour.

ERWEKA VT (Vibration Sieve Shaker)



The Erweka VT operates on the reciprocating cam principle. The vibration intensity can be adjusted by changing the speed of the drive unit. Available sieve sets include:

- **(VS) Sieve set** with 9 sieves 31 cm Ø, according to DAB and Pharm. Europe, made of phosphor bronze.
- **(VS/S) ditto**, sieves made of stainless steel.

- **(PS/S) Sieves set** with 4 stainless steel sieves 20 cm Ø according to DAB 1998, ISO 565/3310-1.

Other sieve sets are available on request.

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AR 403 Attachments

ERWEKA GTE (Pelletizer)



The pelletizer GTE (300 mm o.d.) is attached to the AR 403 via the universal gear UG which enables the operator to adjust the operating angle to the optimum position during operation.

The material can be fed in manually or continuously. The rotary movement of the pan carries the product to the top of the pan and then flows down the pan in an even stream to the base. By addition of the requisite amount

of liquid, this process can be converted into the formation of pellets.

The angle of the pan and the amount of liquid added determines the pellet size. When the pan is full, the pellets will continuously roll over the edge of the pan into a waiting container.

All parts coming into contact with the product are made of stainless steel.

ERWEKA DKE/DKS (Coating Pans)



Coating pans are used for coating tablets. The universal gear is required to attach the coating pans to the motor drive. This allows adjusting the angle of the pan and getting an optimum result from the coating process.

An infrared radiator with a fixing rod can be mounted above the pan to accelerate drying if required.

- **DKE**, stainless steel pan, volume 9.5 l, max. capacity 30%, 275 mm outside diameter.
- **DKS**, stainless steel pan, volume 15 l, max capacity 30%, 350 mm outside diameter.

ERWEKA PT (Polishing Drum)



Polishing drum to polish coated tablets. The drum is covered with interchangeable felt parts.

The cover has a 120 mm o.d. opening for observation of the polishing process and addition of the polishing agent during the production process.

The polishing drum (2 l volume, working capacity max. 40%) is designed for attachment to the drive AR 403 unit via the universal gear UG in a similar manner as the coating pan.

ERWEKA HO/HHO (Homogenizer)



For oil-in-water or water-in-oil emulsions. Mixing is achieved by a stainless steel piston.

The volume of the stainless steel hopper is 1.5 liters. The maximum output is 30 kg/hour.

The HHO is equipped with a heated hopper whereas the HO/UA offers a circulation and filling unit.

ERWEKA FDF (Filling and Dosing Liquids)



The two-station dosing and filling unit is suitable for the drop-free filling of all types of liquids.

The dispense volume is easy to adjust. Quick removal of the pumps and valves for cleaning and sterilisation. The receiving container holder can be easily adjusted to accommodate various sizes. All parts coming into contact with the product are made of stainless steel. Maximum output: 1200 fills/hour.

The EWEKA FDF comes without pumps. Optional dosing pumps for a filling volume of 5 to 30 ml (DP 30) and 20 to 150 ml (DP 150) are available.

The FDF comes with one set of filling nozzles 6 mm and 8 mm o.d.. Filling nozzles with different diameters are available on request.

ERWEKA FDS (Filling and Dosing Ointments etc.)



The dosing machine FDS is suitable for filling precise quantities of ointments, creams, emulsions and similar products into tubes, cans, bottles etc.

The standard unit has a dosing range of 10 – 200 ml. All parts coming into contact with the product are made of stainless steel. Quick removal of the pump and hopper (19 liter volume, maximum capacity 1000 hour)

for cleaning. Filling nozzles of 2, 4 and 8 mm are part of the standard package.

An additional cylinder unit to accommodate the filling range from 0.5 to 25 ml together with dispense heads with different apertures and special heads for filling tubes are available on request.

ERWEKA SG 4/W (Suppository Moulding)



The suppository moulding unit SG 4/W is intended for the production of moulded suppositories, lipsticks and similar moulded products.

The mixing vessel (5 liter volume, working capacity max. 65%) is located in a warm oil bath fitted with a thermostatically controlled heater adjustable between 20°C and 100°C. Stirring, mixing and melting are achieved in

a single operation and are not interrupted during the filling process. The moulds to be filled are positioned on a platform below the dispensing nozzle. A hand wheel enables the operator to move the platform and hence the moulds in both axes.

All parts coming into contact with the product are made of stainless steel. Casting moulds are available on customer's request.

Technical Data All-Purpose Equipment

Unit	Type	Volume max.	Capacity Output max.	UG required	Liquids	Salves / Creams	Powders	Granulates	Pellets	Tablets / Dragees
Ball mill	KM 0	0.5 l	40 %	■			■	■		
	KM 5	5 l	40 %				■	■		
	KM D 5	10 l	40 %				■	■		
	KM 10	10 l	40 %				■	■		
	KM D 10	20 l	40 %				■	■		
Coating pan	DKE	9.5 l	30 %	■						■
	DKS	15 l	30 %	■						■
Cube mixer	KB 15	3.5 l	40 %	■			■	■		
	KB 15 S	3.5 l	40 %	■			■	■		
	KB 20	8 l	40 %	■			■	■		
	KB 20 S	8 l	40 %	■			■	■		
	KB 30	11 l	40 %	■			■	■		
	KB 30 S	11 l	40 %	■			■	■		
Double cone mixer	DKM	11.9 l	50 %				■	■		
Drum hoop mixer	RM 5	5 l	40 %				■	■		
	RM 10	10 l	40 %				■	■		
Dry granulator	TG 2000		20 kg/h					■	■	■
Homogenizer	HO	1.5 l	30 kg/h			■				
Laboatory high speed mixer	SW 1/S	5 l	70 %				■	■		
Laboratory kneader	LK 5	5 l	65 %			■				
Liquid filler	FDF		1200/h		■					
Pelletizer	GTE			■					■	
Planetary mixer	PRS	5 l	60 %		■	■	■			
Polishing drum	PT	2 l	40 %	■						■
Salve filling machine	FDS	19 l	1000/h		■	■				
Suppository moulding machine	SG 4W	5 l	65 %							
Three roller mill	SM		15 kg			■				
	SMS		15 kg			■				
Wet granulator	FGS		30 kg				■	■		
Conical mill	CM		30 kg				■	■		
Y-mixer	YB 5	1.5 l	40 RPM				■	■		

Tablet Press

EP-1



The ERWEKA eccentric single-punch tablet press EP-1 is the ideal tool for R&D and small batch production of round and odd shaped tablets.

The single punch eccentric tablet press EP-1 can manufacture tablets and odd shaped products with a diameter of up to 20mm.

It operates automatically i.e. the tablets are filled, pressed and ejected continuously. The compression force of up to 3 tons and the filling depth of up to 15mm are easily adjusted.

The throughput is maximum 4,000 strokes per hour. All components are either covered with hard chrome, or made of stainless steel AISI 316 or acrylic glass for the front door.

The EP-1 is compliant to GMP and safety regulations.

Technical Data & Options

- Press speed infinitely variable
- Max. tablet diameter 20mm
- Max. filling depth 17mm
- Max. compression force 3t
- Max. tablet output 4000 tablets/hour
- Net weight 135kg
- Dimensions press L/W/H (mm)
425/385/640
- Dimensions with attachment parts
L/W/H (mm) 550/400/800
- Power supply 230V/50-60Hz
- Electrical load 2.2 kVA

Options

- + Measurement and display of compression force
- + Data transfer option
- + Data logging software
- + Standard data export via RS232 interface
- + Support table in stainless steel
- + Single punch and die set for round tablets, 4.0 mm to 20.0 mm diameter
- + Single punch and die set for shaped tablets, 4.0 mm to 20.0 mm diameter
- + Development of STR 1 or STS 1 single punch and die set
- + IQ /OQ documents
- + Transformer for EP-1 (from 110V to 230V)



As a world leader in test equipment for the world wide pharmaceutical and life science industry we design, manufacture, install and service the complete range of test equipment for pharmaceutical research, development and quality control.

- Dissolution testers
- Physical testers
- All-purpose equipment
- R&D and production equipment

ERWEKA GmbH
Ottostr. 20 - 22
63150 Heusenstamm
Germany

+49 6104 6903-0
sales@erweka.com
www.erweka.com

Technical specifications of the described products are stated without obligation and subject to change at any time.

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